RSAAE: Residual Self-Attention-Based Autoencoder for Hyperspectral Anomaly Detection

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***Abstract*— Autoencoder (AE) has been widely used in the field of hyperspectral anomaly detection. It is assumed that the background can be reconstructed well, but the anomalies cannot. Hence, the pixels with larger reconstruction errors are considered anomalies. However, owing to the strong nonlinear representation ability of AE, it is difficult to distinguish between background and anomalies. To address this problem, we propose a residual self-attention-based AE (RSAAE) for hyperspectral anomaly detection. RSAAE consists of dense residual self-attention (RSA) modules, an encoder, and a decoder. First, a novel RSA module is designed, which can effectively extract the main features and weaken the ability of subsequent networks to reconstruct anomalies, as well as preserve the original features to avoid the deterioration of network performance after the use of dense self-attention modules. Furthermore, inspired by manifold learning, we assume that the background is low-rank in the original space, and has the same property in the latent space after dimensionality reduction. We proposed a low-rank loss function to constrain the latent space, thereby suppressing anomaly reconstruction. Experiments on four real hyperspectral image (HSI) datasets showed that the proposed RSAAE method can produce more accurate detection results than eight popular methods.**

***Index Terms*— Autoencoder (AE), hyperspectral anomaly detection, low-rank loss function, reconstruction error, residual self-attention (RSA) module.**

I. Introduction

HYPERSPECTRAL images (HSIs) contain hundreds of narrow spectral bands that can provide rich spatial and spectral information for identifying ground objects. HSIs have been widely used in land cover classification [1], [2], change detection [3], [4], target detection [5], [6], and anomaly detection [7], [8]. Hyperspectral anomaly detection aims to distinguish abnormal targets without prior spectral information. The anomalies are generally characterized by a low probability of occurrence and significant spectral differences with the surrounding background. Existing hyperspectral anomaly detection methods are divided into three main categories: statistic-based, reconstruction-based, and deep learning-based.

The statistical models first use global or local pixels to estimate a background statistical model and then evaluate the anomalies by calculating the distance between the tested pixel and the background statistical model. Pixels with larger distances are regarded as anomalies. As one of the earliest solutions, Reed and Yu [9] proposed a well-known Reed-Xiaoli (RX) method. The RX method uses the mean and covariance matrix of global pixels to estimate the background statistical model. Anomalies are then detected by calculating the Mahalanobis distance between the tested pixels and the estimated background model. Several improved versions were then developed based on the RX method. For example, the local RX (LRX) method [10] uses a dual-window strategy to estimate the local background model. To estimate the background model better, Guo et al. [11] proposed a weighted RX method. Liu and Chang [12] used a multiple-window strategy to adapt to abnormal targets with different scales. Schaum [13] proposed a subspace RX method to reduce the impact of abnormal pollution on background modeling. Kwon and Nasrabadi [14] proposed a kernel RX (KRX) method, which projects the original HSI into a high-dimensional feature space, and makes full use of the nonlinear information of bands. Zhou et al. [15] combined KRX with a clustering technique and proposed a cluster KRX (CKRX) method. Both KRX and CKRX have global and local versions.

The theoretical basis of representation-based methods is that a designed background dictionary can be used linearly to represent tested pixels, and it is assumed that the background can be represented well, but the anomalies cannot. Yuan et al. [16] proposed an anomaly detection method based on local sparse divergence, which calculates the sparse divergence of pixels in each band and considers the pixels with larger dispersion as anomalies. A collaborative representation detector (CRD) [17] uses the dual-window strategy to construct a local dictionary to collaboratively represent the center pixel, and ℓ_2-norm minimization is used to enhance the collaboration ability between adjacent pixels. Low-rank and sparse matrix decomposition (LRSMD) theory is an important branch of representation-based methods, which has been widely used in hyperspectral anomaly detection. Xu et al. [18] introduced low-rank and sparsity-inducing regularization terms to constrain the

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background dictionary representation coefficient and used a residual matrix to distinguish anomalies. Zhang et al. [19] proposed an LRSMD-based Mahalanobis distance method, which used the GoDec algorithm to optimize the model. Qu et al. [20] and [21] employed the LRSMD method to process abundance vectors instead of the original HSI and proposed a mean shift clustering-based dictionary construction method. This improves the discrimination ability for anomalies. Cheng and Wang [22] proposed an anomaly detection method based on a graph and total variation regularized low-rank representation (GTVLRR) model that fully considers the global and local characteristics of HSIs. Subsequently, Cheng and Wang [23] used a union dictionary to represent the background and anomalies separately and proposed a density peak clustering-based dictionary construction strategy. Meanwhile, LRSMD theory is also applicable to tensor decomposition. Li et al. [24] first decomposed the 3-D HSI into low-lank and sparse tensors, then used tucker decomposition to further decompose the low-rank tensor into core tensor, and finally realized anomaly detection by the unmixing method.

In recent years, deep learning theory has been widely used for detecting hyperspectral anomalies. Jiang et al. [25] proposed weakly supervised discriminative learning with a spectral-constrained generative adversarial network (GAN) for hyperspectral anomaly detection, which used density-based spatial clustering of applications with noise (DBSCAN) to construct potential background training samples and introduced a novel Kullback-Leibler divergence-based orthogonal projection divergence spectral constraint. In [26] and [27], DBSCAN and GAN were also used to construct background training samples and learn the background features, respectively. Jiang et al. [28] combined an autoencoder (AE) and Gaussian mixture model and proposed an end-to-end joint-optimized anomaly detection model based on density estimation, which was named E2E-LIADE. Auto-AD [29] used a fully convolutional AE (CAE) with skip connections as a training network, which also introduced a novel adaptive-weighted loss function to further suppress the anomaly reconstruction. Fan et al. [30] introduced a super-pixel segmentation-based graph regularization term into the AE, which could effectively preserve the geometric structure and local spatial consistency of the HSIs. Liu et al. [31] considered the detection of anomalies in the frequency domain and decomposed the HSIs into high- and low-frequency components.

It is not difficult to find that most of the deep learning-based methods are AE-style networks. The existing AE-style-based hyperspectral anomaly detection methods still suffer from the following problems:

1) First, owing to its strong nonlinear representation ability, AE can reconstruct not only the background but also the anomalies. Therefore, the background and anomalies cannot be sufficiently separated when reconstruction error is used as the detection method.
2) Second, as in [25], [26], and [27], semi-supervised models need to first construct coarse background training samples, which have some limitations owing to the problems of abnormal pollution and incomplete background training samples.
3) Finally, most networks consider only the spectral differences between the background and anomalies and ignore the spatial attributes of the original HSI.

To solve the aforementioned problems, this article proposes a residual self-attention-based AE (RSAAE) for hyperspectral anomaly detection. The RSAAE consists of three parts: several residual self-attention (RSA) modules for feature augmentation, an encoder for coding, and a decoder for decoding. Furthermore, we propose a novel RSA module. The purpose of the self-attention mechanism is to highlight the features with higher attention weight values. In HSIs, the proportion between background and anomalies is unbalanced, and the background is dominant. Therefore, the background with larger weight values, and the background features become the main features. However, the original self-attention module used the Softmax function [32] to generate a weight matrix with a weight value range of [0, 1]. Then, the weight matrix is multiplied element-wise by the original feature map. When dense self-attention modules are used, the values of the feature map become smaller, leading to the loss of the original features and changes in network performance. To solve this problem, we introduce a shortcut connection and linear transformation to construct a novel RSA module. The proposed RSA module not only effectively extracts the main features and weakens the ability of the subsequent AE to reconstruct the anomalies, but also preserves the original features. Inspired by manifold learning, it is assumed that the background part is low-rank in the original space, and has similar spatial attributes in the latent space. Based on this, we propose a low-rank loss function to constrain the latent space, which can further suppress anomaly reconstruction.

There are two main contributions of the proposed RSAAE:

1) A novel RSAAE for hyperspectral anomaly detection is proposed. The RSA module is able to highlight background features and also avoid feature variation caused by the use of dense self-attention modules.
2) A novel loss function for a low-rank constraint is proposed, which can preserve the low-rank property of the background in the latent space and suppress the reconstruction ability for anomalies.

The remainder of this article is organized as follows. Related works are briefly described in Section II. Section III describes the RSAAE in detail. Section IV presents experimental results based on four real HSI datasets. In Section V, the proposed method and the experimental results are further discussed. Finally, the article is concluded in Section VI.

II. Problem Formulation and Related Works

In this section, we briefly introduce the basic principles of AE, LRSMD, and residual network (ResNet) [33]. The problems are formalized as follows. Let denote a 3-D HSI as $\mathbb{Y} \in \mathbb{R}^{H\times W\times D}$, where H, W and D represent the height, width, and dimension, respectively. The transformed 2-D form is $\mathbf{Y} = [\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_i, \ldots, \mathbf{y}_l] \in \mathbf{R}^{D\times L}$, where $L = H \times W$ represents the total number of pixels, and $\mathbf{y}_i \in \mathbf{R}^{D\times 1}$ is the ith pixel in $\mathbf{Y}$. Additionally, variable x is used as an instance in (7) and (9), respectively.

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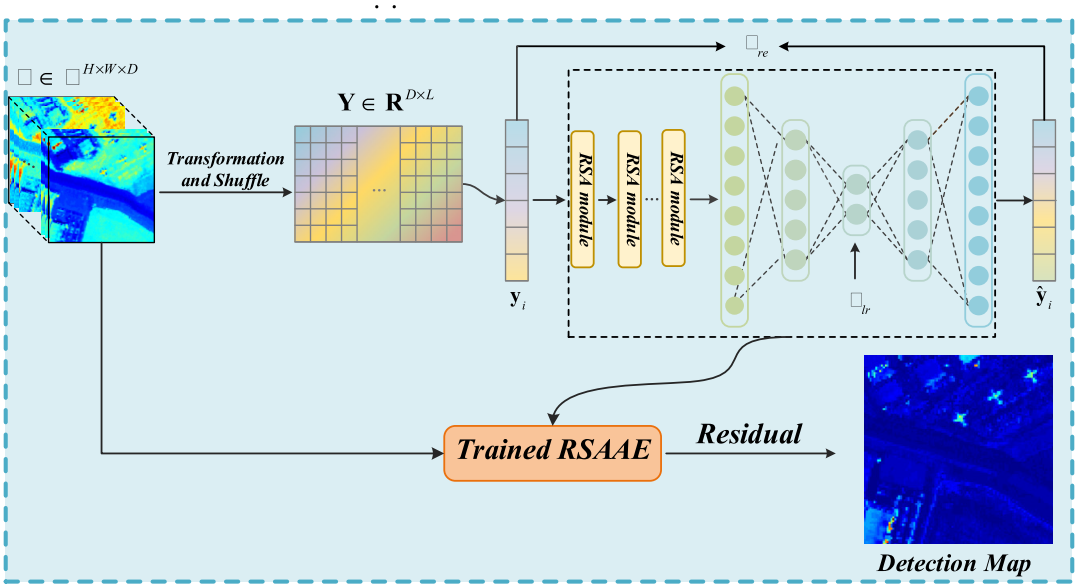


Fig. 1. Overall flowchart of the proposed RSAAE method.

A. Autoencoder

AE is an unsupervised neural network model that consists of an encoder and a decoder. The encoder maps the input data $\mathbf{y}_i$ into a lower dimension space and generates code $\mathbf{z}_i$, which contains important features. The decoder reconstructs the output data $\bar{\mathbf{y}}_i$ from $\mathbf{z}_i$. The encoder and decoder processes can be defined as

$$\mathbf{z}_i = \sigma(\mathbf{W}\mathbf{y}_i + \mathbf{b}) \tag{1}$$

$$\bar{\mathbf{y}}_i = \bar{\sigma}(\bar{\mathbf{W}}\mathbf{z}_i + \bar{\mathbf{b}}) \tag{2}$$

where $\mathbf{W}$ and $\bar{\mathbf{W}}$ are weight matrices, $\mathbf{b}$ and $\bar{\mathbf{b}}$ are biases, and σ and $\bar{\sigma}$ are activation functions. Generally, the distance-based function (such as Euclidean distance and spectral angular distance) is used as a reconstruction loss function to conduct network training through backpropagation.

B. Low-Rank and Sparse Matrix Decomposition

LRSMD-based methods consider the spatial attributes of HSIs, which assume that the background part is low-rank and the abnormal part is sparse. The LRSMD model expresses the original 2-D HSI as the sum of a low-rank matrix $\mathbf{B} = [\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_i, \ldots, \mathbf{b}_l] \in \mathbf{R}^{D\times L}$ and a sparse matrix $\mathbf{S} = [\mathbf{s}_1, \mathbf{s}_2, \ldots, \mathbf{s}_i, \ldots, \mathbf{s}_l] \in \mathbf{R}^{D\times L}$. Meanwhile, the low-rank matrix can be expressed as the product of the background dictionary $\mathbf{A} = [\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_i, \ldots, \mathbf{a}_k] \in \mathbf{R}^{D\times K}$ (where K is the number of dictionary atoms) and the corresponding representation coefficient $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_i, \ldots, \mathbf{x}_l] \in \mathbf{R}^{K\times L}$. The optimization problem of the LRSMD can be written as

$$\min_{\mathbf{X},\mathbf{S}} \|\mathbf{X}\|_* + \lambda\|\mathbf{S}\|_{2,1}, \quad \text{s.t. } \mathbf{Y} = \mathbf{A}\mathbf{X} + \mathbf{S} \tag{3}$$

where $\|\cdot\|_*$ and $\|\cdot\|_{2,1}$ express the nuclear norm and $\ell_{2,1}$-norm, respectively, and λ is a trade-off parameter.

Generally, the alternating direction method of multipliers (ADMM) [34] is used to optimize (3). The ADMM is an important method for solving separable convex optimization problems. Owing to its fast-processing speed and great convergence performance, ADMM has been widely used in statistical learning and machine learning. After the optimization of (3) is completed, the anomalies are identified by calculating the ℓ_2-norm of the column vector of $\mathbf{S}$. Specifically, the anomaly score of the ith pixel is defined as $\|\mathbf{s}_i\|_2$. When the anomaly score is larger than a predefined threshold, the pixel is identified as an anomaly.

C. Residual Network

In 2016, He et al. [33] first proposed ResNet, which can be regarded as the sum of identity mapping and residual mapping. The output of ResNet is expressed as

$$\text{ResNet}(\mathbf{y}_i) = \mathbf{y}_i + F(\mathbf{y}_i) \tag{4}$$

where $F(\cdot)$ is a residual mapping function. Identity mapping is implemented using a shortcut connection. To some extent, ResNet alleviates the problems of performance degradation and gradient explosion that occur with the increase of network layers.

III. Methods

Fig. 1 shows the overall flowchart of the proposed RSAAE. The RSAAE consists of dense RSA modules, an encoder, and a decoder. In the training phase, the shuffle function was used to disrupt the sequence of samples. The detection phase does not need to be out of sequence.

A. Proposed RSA Module

The self-attention mechanism [35] is the core of the transformer and has been applied to remote sensing image processing tasks, including image classification [36], change detection [37], and denoising [38]. The self-attention mechanism aims to capture the correlation between variables and highlight important features. The self-attention mechanism can be regarded as a mapping model between a query ($\mathbf{Q}$) and

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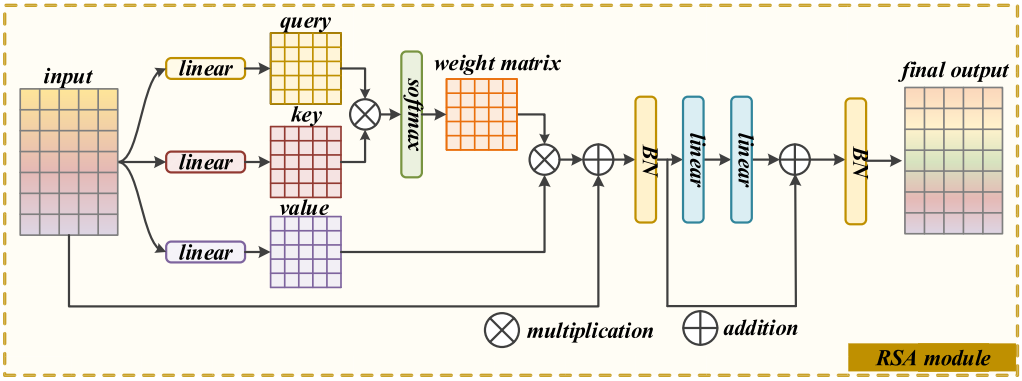


Fig. 2. Overall flowchart of the proposed RSA module.

a key (**K**)-value (**V**) pair. Taking the global sample as an example, it can be regarded as

$$\mathrm{SA}(\mathbf{Y}) = \mathrm{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_K}}\right)\mathbf{V} \tag{5}$$

in (5), **Q**, **K**, and **V** are obtained by a linear transformation

$$\begin{aligned}\mathbf{Q} &= \mathbf{W}_q\mathbf{Y}\\ \mathbf{K} &= \mathbf{W}_k\mathbf{Y}\\ \mathbf{V} &= \mathbf{W}_v\mathbf{Y}\end{aligned} \tag{6}$$

where $\mathbf{W}_q$, $\mathbf{W}_k$, and $\mathbf{W}_v$ are the weight matrices. The product of **Q** and $\mathbf{K}^{\mathrm{T}}$ was used to calculate the weight matrix, and the Softmax function was used to normalize the weight value in the range [0, 1]. It is worth noting that to avoid the value of the attention matrix becoming too large, the scaled dot-product model is used to adjust the weight values. The dot-product result is divided by the square value of the feature dimension of **K**. Finally, the product of the weight matrix and **V** is the new feature map of the current node.

As the Softmax function normalizes the value of the weight matrix in the range [0, 1], the output response of the new feature map becomes smaller. In particular, after using dense self-attention modules, the value of the output feature map becomes extremely small. As a result, the original features are lost, and the network performance changes. To solve this problem, we draw on the concept of ResNet and propose a novel RSA module.

To better fuse the original and new feature maps, shortcut connection and fully connected layers are introduced into the RSA module. The proposed RSA module is shown in Fig. 2. It can be clearly seen that after using the first shortcut connection, the feature maps are normalized by the batch normalization (BN) layer [39]. The BN layer can accelerate network convergence and prevent overfitting when a smaller learning rate is used. The BN process can be expressed as

$$\mathrm{BN}(x) = \beta\left(\frac{x - \mu_{\mathrm{BN}}}{\sqrt{\sigma^2_{\mathrm{BN}} + \varepsilon}}\right) + \eta \tag{7}$$

where μ_{BN} and σ^2_{BN} are the mean and variance of each batch. β and η are the hyperparameters that need to be determined by learning. ε is a bias, which can be set to a smaller real number to avoid the denominator being equal to 0. In this article, ε was set to 0.01. Subsequently, the two fully connected layers and the shortcut connection are used to realize the second identity mapping. The result that passes through the second shortcut connection is inputted to the BN layer to obtain the final output. The output of the proposed RSA module can be represented as

$$\mathrm{RSA}(\mathbf{y}_i) = \mathrm{BN}(f_c(\mathrm{BN}(\mathbf{y}_i + \mathrm{SA}(\mathbf{y}_i))) + \mathrm{BN}(\mathbf{y}_i + \mathrm{SA}(\mathbf{y}_i))) \tag{8}$$

where f_c represents the fully connected layers and BN is the BN layer. Using dense RSA modules can not only effectively extract main features, but also improve the depth and stability of the network.

B. Proposed RSAAE

The RSAAE is an AE-style unsupervised network. It is assumed that there are obvious spectral differences between the background and anomalies. Moreover, the proportion of anomalies in the HSI scene is small. An AE-style network can learn the dominant feature, that is, the background feature in the HSI scene. The background can be reconstructed well, but the anomalies cannot. Therefore, in this article, the reconstruction error was adopted as the detection method.

In RSAAE, dense RSA modules are first used to enhance the background feature. The encoder and decoder are composed of fully connected layers that include several neurons and activation functions. Since the HSI is normalized before training, the pixel values are in the range [0, 1]. To avoid negative values, we use the Sigmoid function [40] as the activation function of the encoder and decoder, which is defined as

$$\mathrm{Sigmoid}(x) = \frac{1}{1 + e^{-x}}. \tag{9}$$

Notably, the last layers of the encoder have no activation function. In this article, the number of RSA modules is set to 9, and the sizes of $\mathbf{W}_q$, $\mathbf{W}_k$, and $\mathbf{W}_v$ are set as the dimension of the original HSI, that is, D. z represents the dimension of the latent space.

The decoder outputs the reconstructed HSI $\bar{\mathbf{Y}} \in \mathbf{R}^{D\times L}$. To avoid the interference of anomalies in the model and enhance the robustness of the model, we use smooth ℓ_1-norm [41] as the reconstruction loss,

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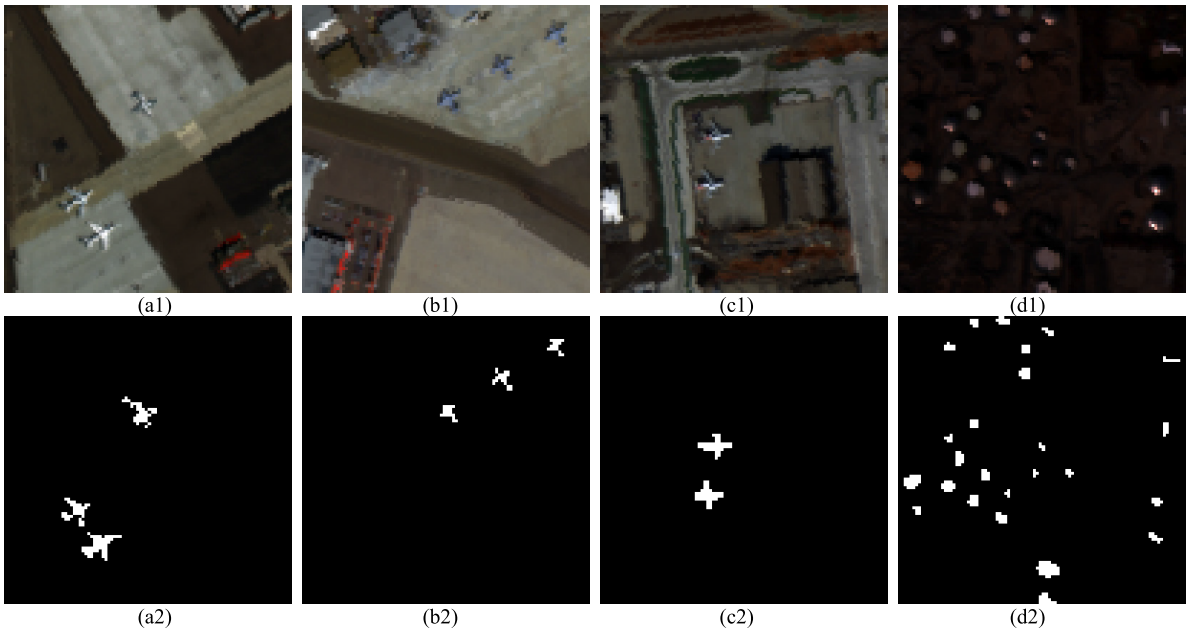


Fig. 3. Four datasets used in the experiments. (a)–(d) Four real HSI datasets. (a1)–(d1) False-color images (bands 37, 18, and 8 as RGB) and (a2)–(d2) Corresponding ground-reference maps of anomalies.

which can be regarded as

$$\mathcal{L}_{\text{re}} = \begin{cases} 0.5(\mathbf{Y} - \bar{\mathbf{Y}})^2, & \text{if } \left|\mathbf{Y} - \bar{\mathbf{Y}}\right| < 1 \\ \left|\mathbf{Y} - \bar{\mathbf{Y}}\right| - 0.5, & \text{otherwise.} \end{cases} \tag{10}$$

Inspired by manifold learning, we assume that the background part is low-rank in the original space, and the coding in the latent space also has similar properties. Thus, we constrained latent coding with the nuclear norm to maintain the low-rank property of the background. The function used to calculate the nuclear norm is as follows:

$$\|\mathbf{Z}\|_* = \mathbf{tr}(\sqrt{\mathbf{Z}^T\mathbf{Z}}) \tag{11}$$

where tr(·) represents the sum of the eigenvalues of the matrix, that is, the sum of the diagonal elements. However, the magnitude difference between the nuclear norm and reconstruction loss is too large, which causes the nuclear norm to drop rapidly during the training process and affects the stability of the network. Thus, we used the logarithm of the nuclear norm as the low-lank loss function

$$\mathcal{L}_{\text{lr}} = \ln(\|Z\|_*). \tag{12}$$

The total loss function of the model is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{re}} + \alpha\mathcal{L}_{\text{lr}} \tag{13}$$

where α is a trade-off parameter that is set to adjust the contribution of the low-rank loss function in the network. The final detection map was calculated using the reconstruction error. Distance-based methods are typically used, such as spectral angular distance, ℓ_1-norm and ℓ_2-norm. In this article, we used ℓ_2-norm to calculate the anomaly score of the ith pixel

$$s_i = \left\|\mathbf{y}_i - \bar{\mathbf{y}}_i\right\|_2. \tag{14}$$

Table I lists the architectural features of the proposed RSAAE model. The main parameters of the RSA module are the sizes of **Q**, **K**, and **V**, which are uniformly set to the spectral dimension D of the original HSI, and the sizes of the two fully connected layers in the RSA module are (D, D).

TABLE I
ARCHITECTURE FEATURES OF THE PROPOSED RSAAE MODEL

RSA Modules		
Layers	Type	Parameters
1	RSA module	D
…	RSA module	D
9	RSA module	D
Encoder		
Layers	Type	Parameters
1	Linear+*Sigmoid*	(D, 1000)
2	Linear+*Sigmoid*	(1000, 1000)
3	Linear+*Sigmoid*	(1000, 1000)
4	Linear	(1000, 7)
Decoder		
Layers	Type	Parameters
1	Linear+*Sigmoid*	(7, 1000)
2	Linear+*Sigmoid*	(1000, 1000)
3	Linear+*Sigmoid*	(1000, 1000)
4	Linear+*Sigmoid*	(1000, D)

IV. EXPERIMENTS

A. Experimental Setup

1) HSI Datasets: Four real HSI datasets were used in the experiments. Table II details the main characteristics of these datasets, including the captured location, sensor, spatial size, the number of bands, spatial resolution, abnormal type, and the number of abnormal pixels. Fig. 3 shows the false-color images and corresponding ground-reference maps of the four datasets.

2) Benchmark Methods: To validate the effectiveness of the proposed method, we compared RSAAE with three categories of benchmark methods. These are statistic-based methods, i.e., RX method [9], LRX method [10], and CKRX method (local version) [15]; representation-based methods, i.e., CRD [17]

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TABLE II
SOME IMAGE CHARACTERISTICS OF THE FIVE DATASETS

Datasets	Location	Sensor	Spatial size	No. of Bands	Spatial resolution	Abnormal Type	No. of Abnormal Pixels
Dataset I	Sandiego Airport	AVIRIS	100×100	189	3.5 m	Aircraft	134
Dataset II	Sandiego Airport	AVIRIS	100×100	189	3.5 m	Aircraft	58
Dataset III	Los Angeles Airport	AVIRIS	100×100	205	7.1 m	Aircraft	87
Dataset IV	Texas Coast Urban	AVIRIS	100×100	204	17.2 m	Building	155

and GTVLRR [22]; and deep learning-based methods, i.e., AE, E2E-LIADE [28], and Auto-AD [29].

3) Evaluation Indices: To assess detection performance, this article considered the following four indices: 2-D receiver operating characteristic (ROC) curves [42], 3-D ROC curve [43], area under the ROC curve (AUC) [44], and box-whisker plot [45].

Two versions of 2-D ROC curves were used in the experiments: ROC (FPR, TPR) and ROC (τ, FPR). ROC (FPR, TPR) describes the relationship between the false positive rate (FPR) and the true positive rate (TPR) under different thresholds τ. The closer the curve is to the upper-left corner of the coordinate axis, the better the performance. With a certain τ, the formulas for calculating FPR and TPR are as follows:

$$\mathrm{FPR} = \frac{\mathrm{FP}}{\mathrm{FP}+\mathrm{TN}} \tag{15}$$

$$\mathrm{TPR} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \tag{16}$$

where TP and TN represent the numbers of detected true anomalies and background pixels, respectively. FP and FT represent the number of pixels falsely detected as anomalies and the background, respectively. Additionally, ROC (τ, FPR) can reflect the false alarm rate with different thresholds, and the ideal performance is that the curve is close to the lower-left corner of the coordinate axis. The AUC value is the AUC, which can quantitatively describe the accuracy of the different anomaly detection methods.

Box-whisker can visually reflect the separability between anomalies and background. The larger the gap between the anomaly box and the background box, the greater the separation. The smaller the background box, the better the suppression of the false alarm rate.

4) Parameter Settings: In addition to the RX method, several parameters must be set in the proposed method and the remaining benchmark methods. In the experiments, the dual-window sizes of LRX were set to (3, 23), (9, 13), (5, 7), and (9, 13) for datasets I–IV, respectively. The k-means method was used as the clustering method in CKRX, and the k values were set to 15 for datasets I and II, 20 for dataset III, and 5 for dataset IV. The dual-window sizes were set to (5, 19), (3, 23), (3, 7), and (3, 7) for datasets I–IV, respectively. Additionally, the trade-off parameter λ was set to 10^{-4} for all datasets. The dual-window sizes of the CRD were set to (17, 21), (19, 21), (11, 13), and (17, 19) for datasets I–IV, respectively. The trade-off parameter λ was set to 10^{-6}. Referring to [18], the trade-off parameters of the GTVLRR were $\lambda = 0.05$, $\beta = 0.2$, and $\gamma = 0.02$. In addition to the absence of the RSA modules and low-rank loss function, the remaining hyperparameters of the AE and RSAAE are the same. The learning rate and the number of training iterations were 9×10^{-5} and 80, respectively, and the batch sizes for the four datasets were set to 1024, 128, 256, and 256, respectively. Additionally, the trade-off parameter α in RSAAE was set to 0.1 for dataset III, and those of the remaining datasets were set to 0.01. In Auto-AD, the learning rate, the number of training iterations, and the threshold value were set to 0.01, 1000, and 1.5×10^{-5}, respectively. The hyperparameters of E2E-LIADE were set as $\lambda_1 = 3$, $\lambda_2 = 0.1$ and $\lambda_3 = 10^{-4}$, respectively.

5) Experimental Organization: The experiments consisted of three parts. Section IV-B presents anomaly detection results based on the four datasets. Section IV-C analyzes the sensitivity of the proposed RSAAE to the hyperparameters. The influence of different self-attention modules is analyzed in Section IV-D. Section IV-E analyzes the effect of the low-lank loss function on the experimental results.

B. Detection Results

Fig. 4 shows the detection maps based on the nine anomaly detection methods for the four datasets. Compared with the eight benchmark methods, RSAAE can accurately detect anomalies. Taking dataset I as an example, LRX, CRD, and Auto-AD cannot clearly detect the positions of abnormal aircraft. Although RX and E2E-LIADE can detect the positions of abnormal aircraft, they cannot fully separate anomalies from the background. CKRX, GTVLRR, and AE can detect anomalies well, but have higher false alarm rates. The experimental results of dataset II are similar to those of dataset I. Additionally, for dataset III, most benchmark methods cannot preserve the shapes of abnormal aircraft. Only AE and RSAAE can achieve this, and RSAAE can better suppress the complex background. For dataset IV, except for LRX, the other methods can detect almost anomalies, but RSAAE still performs well. Although CKRX can highlight the anomalies, the background cannot be fully suppressed. As shown in Fig. 4(a4)–(h4), RSAAE can detect all anomalies and suppress the background.

Fig. 5 shows the 2-D and 3-D ROC curves for the four datasets based on the nine methods. As shown in Fig. 5(a1)–(a4), for the four datasets, the 2-D ROC (FPR, TPR) curves of RSAAE are located in the top left corner, which means that RSAAE has a higher detection rate. As shown in Fig. 5(b1)–(b4), which shows the 2-D ROC (τ, FPR) curves. Although RSAAE has a higher false alarm rate than the statistical methods (RX and LRX), it has a better ability to suppress the false alarm rate than CKRX, AE, and GTVLRR. Meanwhile, the 3-D ROC curves in Fig. 5(c1)–(c4) show that RSAAE has a greater performance for different datasets. In Fig. 6, the box-whisker plot is used to verify the ability of the nine methods to separate the background and

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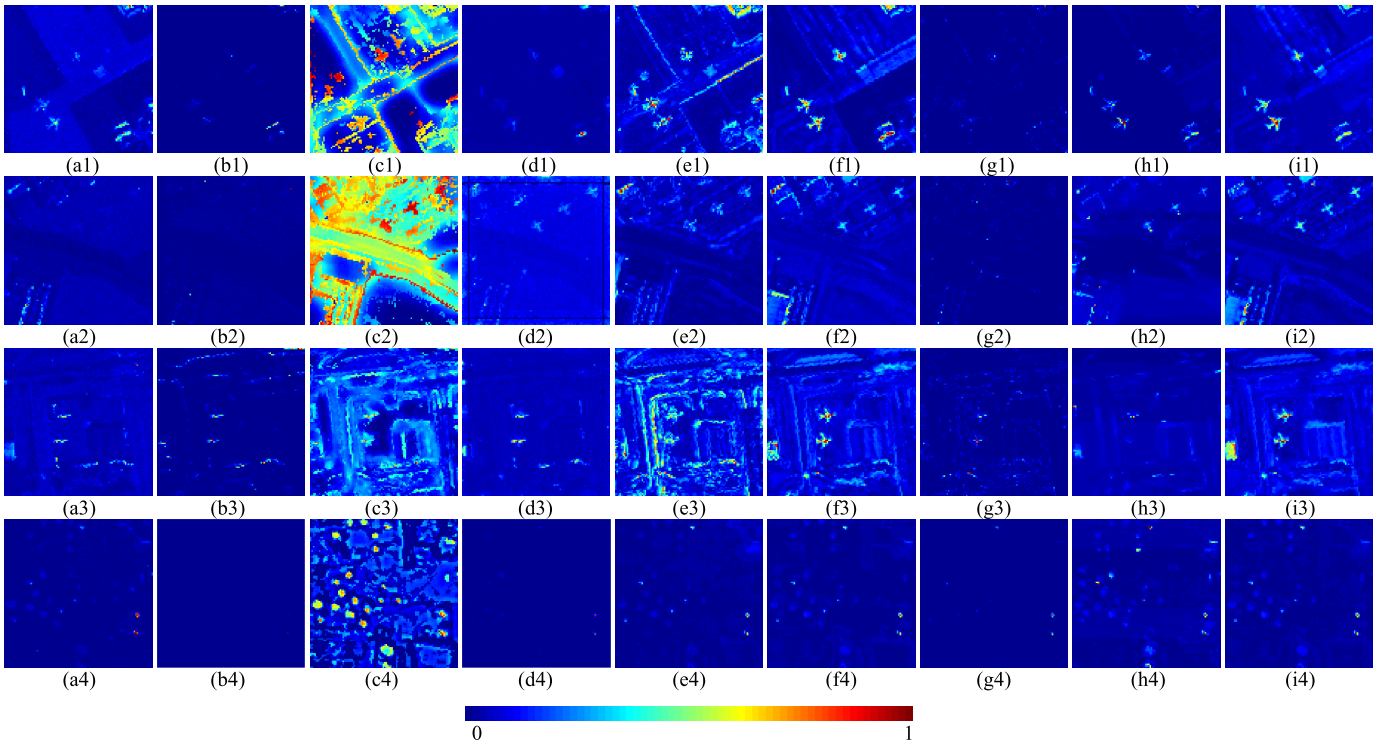


Fig. 4. Detection maps based on nine methods for the four HSI datasets. (a)–(i) Detection maps for RX, LRX, CKRX, CRD, GTVLRR, AE, Auto-AD, E2E-LIADE, and RSAAE, respectively. (a1)–(i1)–(a4)–(i4) Datasets I–IV, respectively.

TABLE III

AUC (FPR, TPR) VALUES OF THE NINE METHODS FOR THE FOUR DATASETS

	GRX	LRX	CKRX	CRD	GTVLRR	AE	Auto-AD	E2E-LIADE	RSAAE
Dataset I	0.9403	0.9417	0.9334	0.9664	0.9311	0.9796	0.8931	<u>0.9846</u>	**0.9920**
Dataset II	0.8885	0.7590	0.9721	0.9865	0.9800	0.9853	0.8568	<u>0.9910</u>	**0.9928**
Dataset III	0.8221	0.8845	0.9026	0.8089	<u>0.9555</u>	0.9495	0.8984	0.8703	**0.9803**
Dataset IV	0.9526	0.9273	0.9908	0.9628	0.9676	0.9871	<u>0.9911</u>	0.9837	**0.9931**

TABLE IV

AUC (τ, FPR) VALUES OF THE NINE METHODS FOR THE FOUR DATASETS

	GRX	LRX	CKRX	CRD	GTVLRR	AE	Auto-AD	E2E-LIADE	RSAAE
Dataset I	0.0588	**0.0069**	0.2423	0.0267	0.0781	0.0562	<u>0.0087</u>	0.0124	0.0559
Dataset II	0.0379	<u>0.0140</u>	0.4433	0.0934	0.0485	0.0636	**0.0087**	0.0375	0.0520
Dataset III	0.0516	**0.0151**	0.1638	0.0537	0.1396	0.0964	<u>0.0165</u>	0.0373	0.0889
Dataset IV	0.0111	**0.0050**	0.0981	<u>0.0051</u>	0.0157	0.0139	0.0056	0.0214	0.0157

anomalies. The background and anomaly boxes of the RSAAE have larger distances and smaller overlaps, which shows better generalization for the four datasets.

The AUC values based on the nine methods are listed in Tables III and IV, where the first and second most accurate results of the AUC (FPR, TPR) and AUC (τ, FPR) for each dataset are highlighted in boldface and underlined, respectively. As shown in Table III, the AUC (FPR, TPR) values of RSAAE are the largest compared with those of the other methods. Taking dataset III as an example, the AUC (FPR, TPR) value of RSAAE is 0.9803, which is 0.1582, 0.0958, 0.0777, 0.1714, 0.0248, 0.0308, 0.0819 and 0.1100 larger than that of RX, LRX, CKRX, CRD, GTVLRR, AE, Auto-AD and E2E-LIADE, respectively. Tables III and IV show that LRX and Auto-AD have smaller AUC (FPR, TPR) and AUC (τ, FPR) values. Although the AUC (τ, FPR) value of RSAAE is not the smallest one, it is smaller than that of AE for most datasets. Specifically, for datasets I-III, the AUC (FPR, TPR) values of RSAAE are 0.0559, 0.0520, and 0.0889, respectively, which are 0.0003, 0.0116, and 0.0075 smaller than those of AE. Table V lists the running times based on the nine methods for the four datasets. The first and second shortest running times are highlighted in boldface and underlined, respectively. It was demonstrated that deep learning-based methods are more time-consuming. Compared with AE, RSAAE requires a longer time, mainly because dense RSA modules increase

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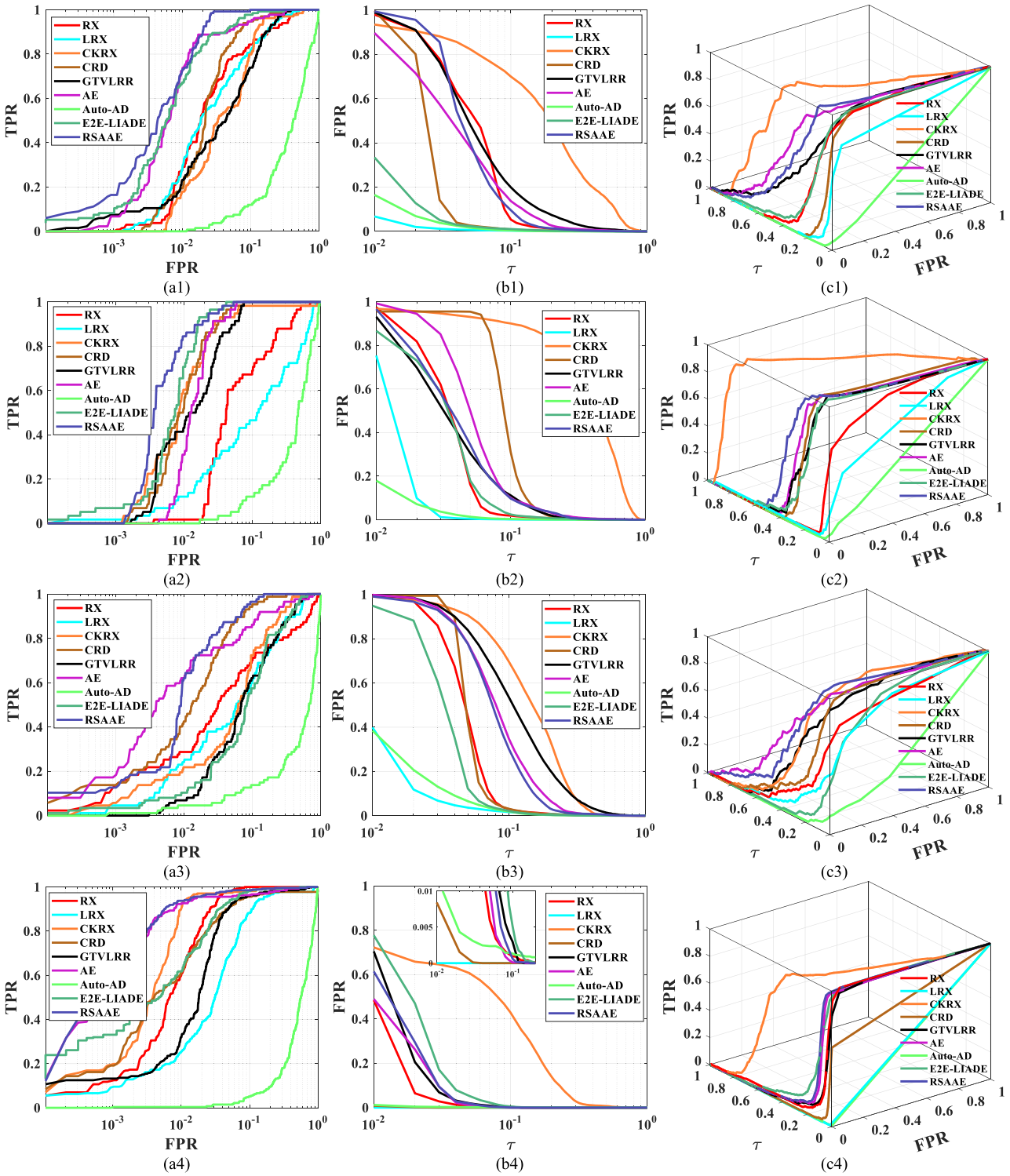


Fig. 5. ROC curves of the nine methods for the four real datasets. (a) 2-D ROC curves of (FPR, TPR). (b) 2-D ROC curves of (τ, FPR). (c) 3-D ROC curves. (a1)–(c1)–(a4)–(c4) Results for datasets I–IV, respectively.

the depth of the network. Also, because of the kernel function used, CKRX results are more time-consuming when there is a large difference between the inner and outer windows, as in the case of datasets I and II.

C. Hyperparameter Analysis

In this section, we analyze some important hyperparameters of the proposed RSSAE to verify their sensitivity. Fig. 7 shows the impacts of the different hyperparameters on the detection results for the four datasets, which include the batch size, the tradeoff parameter α, and the number of RSA modules. As shown in Fig. 7(a), with the increase in batch size from 32 to 256, the AUC values of most datasets (except dataset II) increased. When the batch size ranged from 256 to 2048, the AUC values of the four datasets did not change significantly. The changes in the AUC values of

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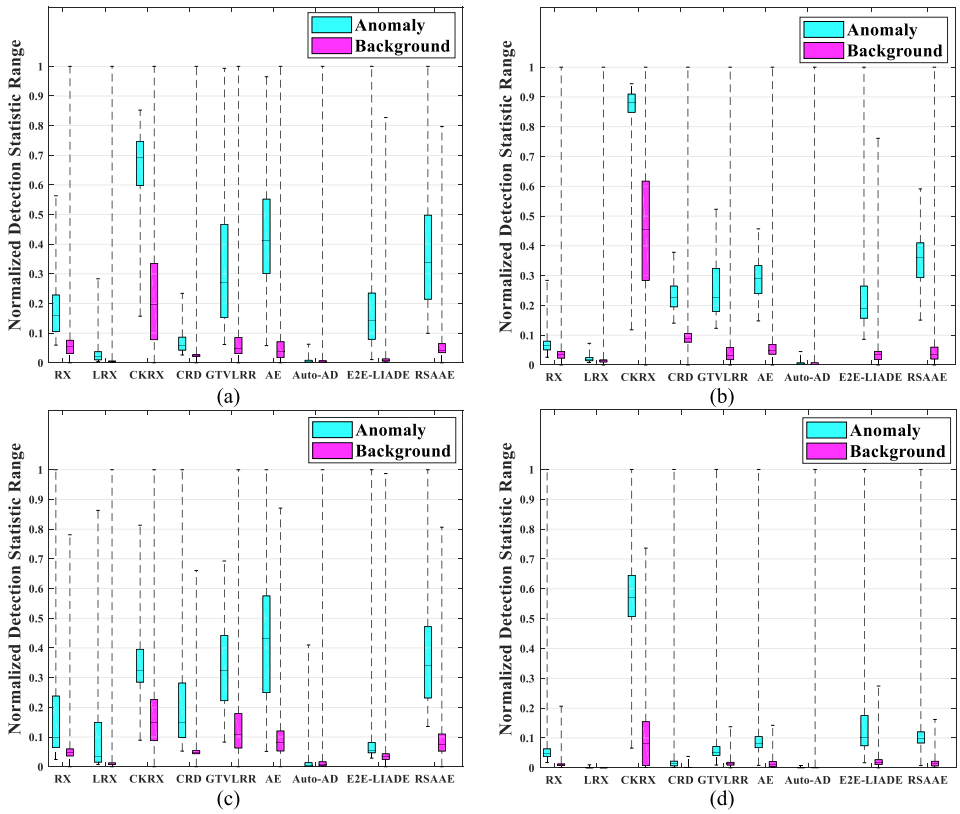


Fig. 6. Box-whisker plots for the nine anomaly detection methods for the four datasets. (a) Dataset I. (b) Dataset II. (c) Dataset III. (d) Dataset IV.

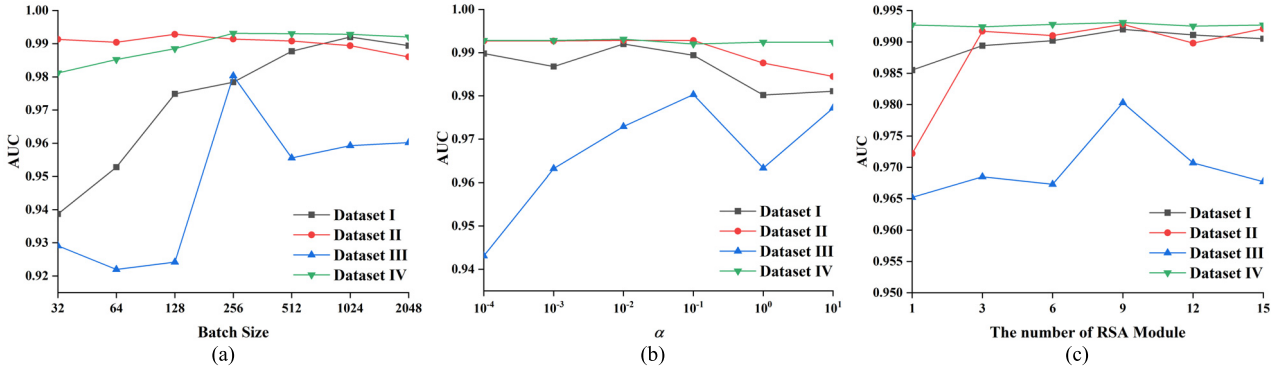


Fig. 7. Impacts of the different hyperparameters on the detection results for the four datasets. (a) Impact of batch size. (b) Impact of the tradeoff parameter α. (c) Impact of the number of RSA modules.

TABLE V
RUNNING TIME (IN SECONDS) OF THE NINE METHODS FOR THE FOUR DATASETS

	GRX	LRX	CKRX	CRD	GTVLRR	AE	Auto-AD	E2E-LIADE	RSAAE
Dataset I	**1.5587**	49.7117	898.7621	11.3878	181.9313	164.4337	233.6808	4621.9743	784.5235
Dataset II	**0.0379**	34.4299	1358.9718	33.3282	191.7638	336.2478	181.3562	4248.4059	728.9580
Dataset III	**0.0516**	33.2499	48.9864	28.7961	199.7956	225.1468	203.0788	8184.7821	614.6658
Dataset IV	**0.0111**	39.8500	20.2857	10.2337	148.4368	219.8072	41.6507	3940.5272	531.6045

the four datasets are 0.0136, 0.0054, 0.0247, and 0.0010, respectively. Additionally, as shown in Fig. 7(b), a larger α value can produce better detection performance. For example, for a range of α from 10^{-4} to 10^{-1}, all four datasets produce the best detection results at 10^{-2} or 10^{-1}. Meanwhile, when the α value continues to increase to 10, the AUC values show a decreasing trend for most datasets. Finally, Fig. 7(c) shows the effect of the number of RSA modules on the detection results. It was demonstrated that with an increase in the number of RSA modules, the performance of the network becomes more stable; however, larger numbers are not always better. Taking dataset I as an example, when the number of RSA

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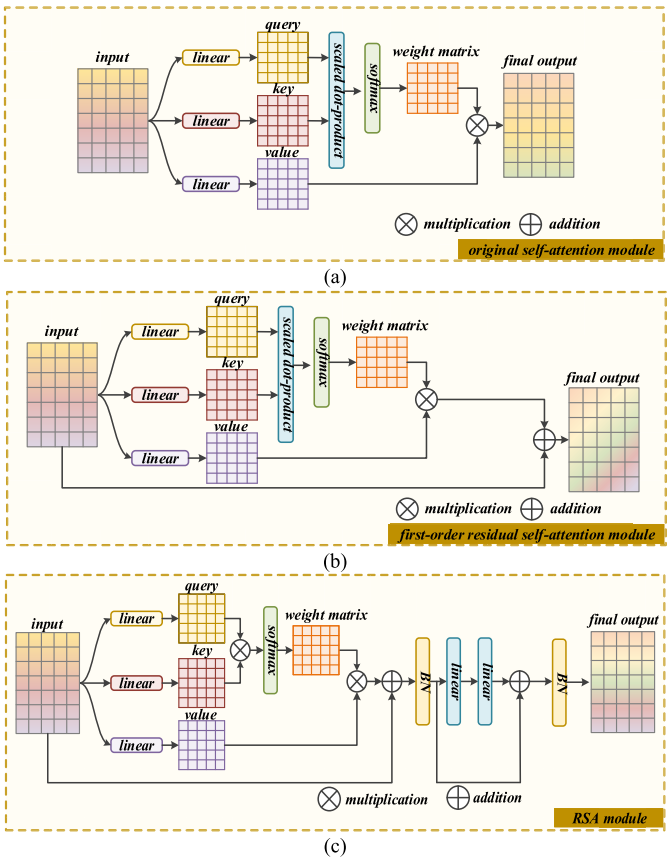


Fig. 8. Three types of self-attention modules. (a) Original self-attention module. (b) First-order RSA module. (c) RSA module.

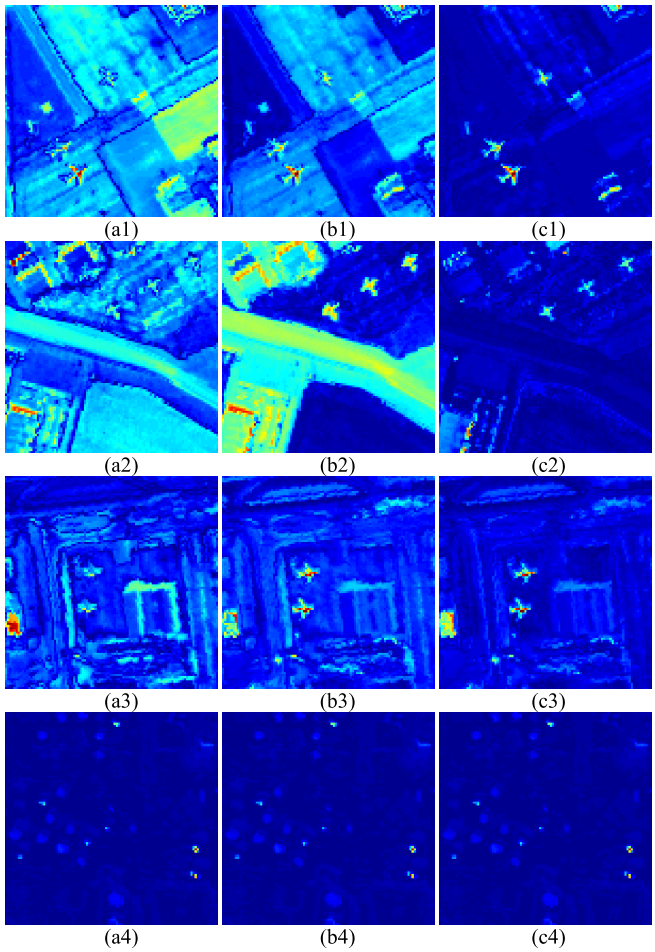


Fig. 9. Detection maps based on different self-attention modules for the four HSI datasets. (a)–(c) Detection maps for the original self-attention module, the first-order RSA module, and the proposed RSA module, respectively. (a1)–(c1)–(a4)–(c4) Results for datasets I–IV, respectively.

modules increases from 1 to 9, the change in the AUC value is 0.0065, while is only 0.0015 from 9 to 15. When the number of RSA modules was 9, 12, and 15, the AUC values were 0.9920, 0.9911, and 0.9905, respectively. In other words, the proposed RSAAE has good robustness to different values of these parameters.

D. Comparison of Different Self-Attention Modules

In this section, we analyze the effect of different self-attention modules on the detection results. Fig. 8 shows three types of self-attention modules, including the original self-attention module, the first-order RSA module, and the proposed RSA module. Under the condition that the subsequent AE and other hyperparameters are consistent, Fig. 9 shows the detection maps of different self-attention modules for the four datasets. It can be seen that the RSA module can produce a higher detection rate and lower false alarm rate. Taking dataset I as an example, neither the first-order residual nor the original self-attention module can accurately detect anomalous aircraft and suppress the background. In particular, using the original self-attention module, the background, and anomalies are almost inseparable. Another example is dataset III: compared with the original self-attention module, first-order RSA and RSA modules can produce a higher detection rate, but the RSA module still has a lower false alarm rate. The main reason is the original self-attention module results in a very small value in the feature map, thus losing the original information. Fig. 10 shows the ROC curves of different self-attention modules for the four HSI datasets, which is consistent with the results of the detection maps.

Taking dataset III as an example, Fig. 11 shows the corresponding attention map of the RSA module, calculated by ℓ_2-norm of attention weight value for each pixel. It can be clearly seen that the weight values of most backgrounds are larger, while that of the two abnormal aircraft are smaller (marked with white circles). This indicates that the RSA module can highlight the features which are dominant in proportion, i.e., the background. Additionally, as shown in Fig. 12, the detection map created by the intermediate HSI after the RSA module enhanced the background reconstruction to a certain extent. Compared with the detection maps created by AE and RSAAE, the intermediate detection maps can better suppress complex backgrounds, such as the areas marked with white circles at the top of the detection maps.

E. Performance of Low-Rank Loss

In this section, we examine the effect of the proposed low-rank loss function. Fig. 13 shows the AUC values with (W) and without (W/O) low-rank loss for the four datasets. It is demonstrated that with the low-lank loss function, the four datasets have larger detection rates. Specifically, compared to the case without the low-rank loss, the AUC value gains are 0.0068, 0.0040, 0.0078, and 0.0003, respectively. Additionally,

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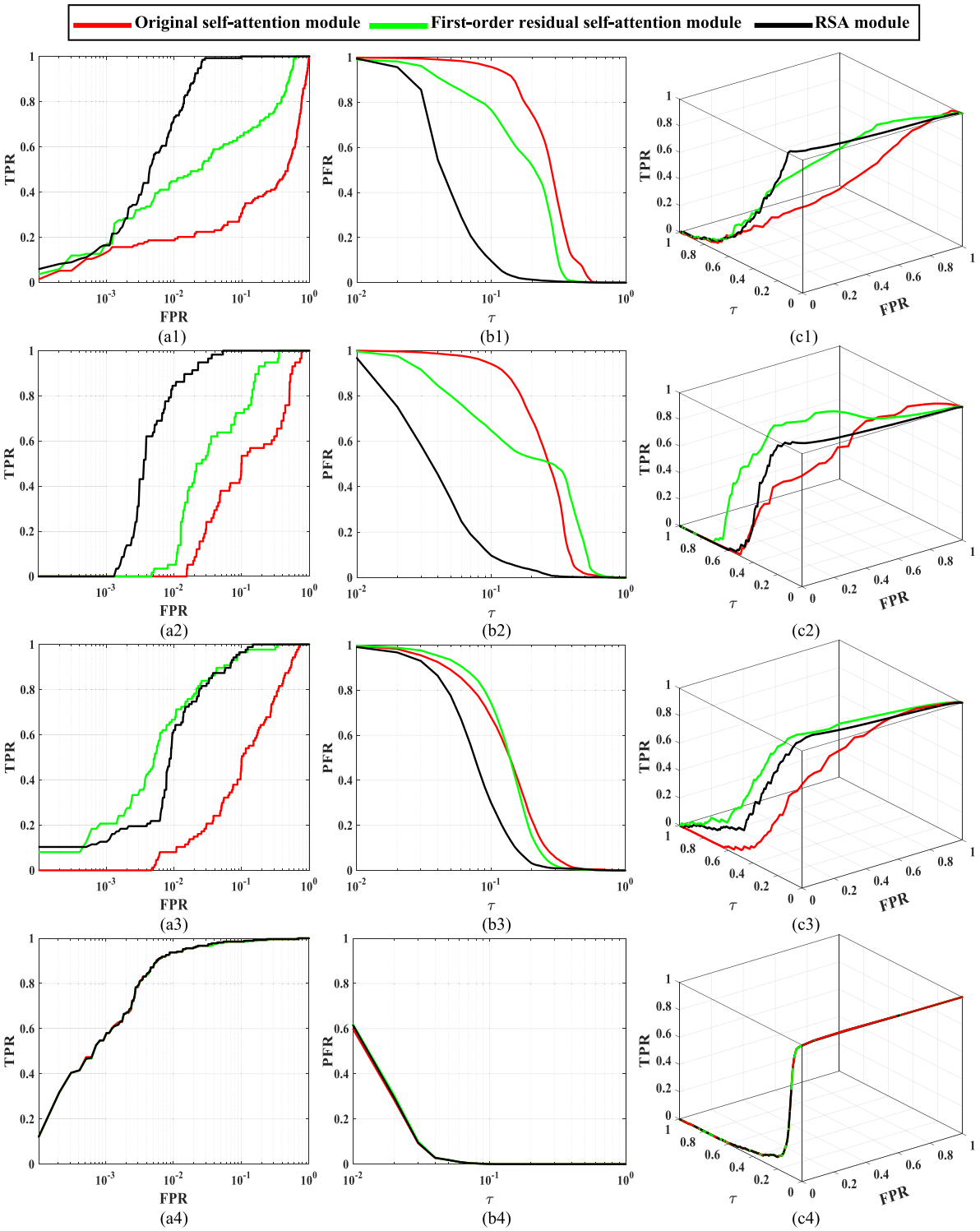


Fig. 10. ROC curves of different self-attention modules for the four HSI datasets. (a) 2-D ROC curves of (FPR, TPR). (b) 2-D ROC curves of (τ, FPR). (c) 3-D ROC curves. (a1)–(c1)–(a4)–(c4) Results for datasets I–IV, respectively.

as shown in Fig. 14, the detection accuracy with low-rank loss is higher. For example, in Fig. 14(c1), the shape of the abnormal aircraft can be completely preserved, but this is not the case in Fig. 14(c2). Similar results were obtained for datasets I and II.

V. Discussion

Deep learning-based hyperspectral anomaly detection has received extensive attention in recent years. Owing to the lack of prior information, much more efforts need to be devoted to it. At present, most deep learning-based anomaly detection methods use AE or CAE as the model backbone and reconstruction error as the detection method. In this article, the novel RSA module acts on the front end of the AE, which can enhance background reconstruction and inhibit anomaly reconstruction. Whether the proposed module can be applied to other HSI processing fields, such as supervised target detection and change detection, is an interesting question that needs

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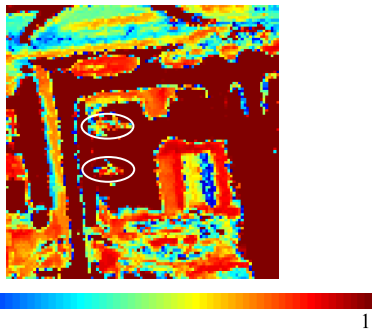


Fig. 11. Visualization of attention map for dataset III.

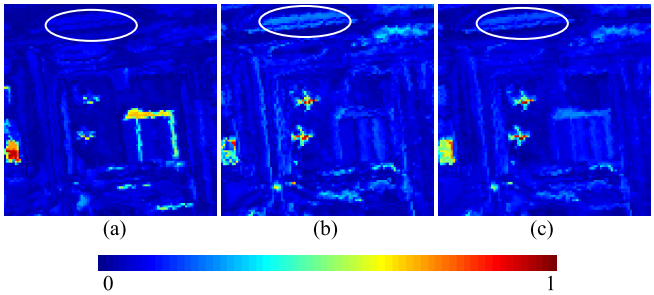


Fig. 12. Detection maps of different stages. (a) Detection map after RSA module. (b) Detection map by AE. (c) Detection map by RSAAE.

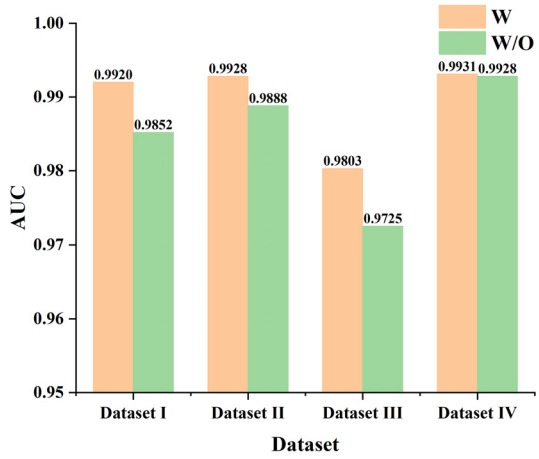


Fig. 13. AUC values of different detection results W and W/O the low-rank loss for the four datasets.

to be answered in future research. Additionally, as shown in Fig. 4 and Table IV, although RSAAE had a higher detection rate and lower false alarm rate than AE, it still had a larger false alarm rate than LRX, CRD, and E2E-LIADE. In future work, we should not only focus on improving the detection rate but also on suppressing the false alarm rate.

When the spectral difference between the background and anomalies is small, such as in dataset IV, the brightness difference in the detection map is not obvious; that is, the background and anomalies are difficult to distinguish. The use of deep learning-based methods to increase the spectral difference between the background and anomalies is a problem worthy of attention. For example, Li et al. [24] proposed an adaptive weighted loss function. The weighted loss is calculated using the reconstruction error and fed back to the network. It is considered that the anomalies have a larger reconstruction error; therefore, a smaller penalty is imposed

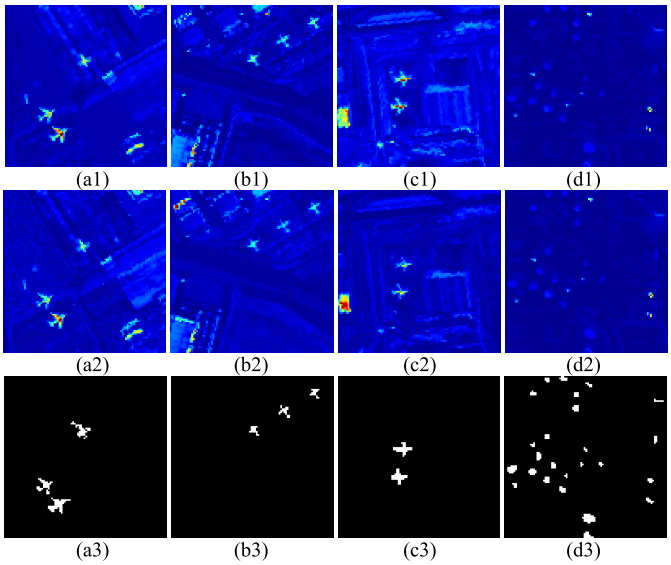


Fig. 14. Detection maps of RSAAE with and without low-rank loss for the four datasets. (a) Dataset I, (b) dataset II, (c) dataset III, and (d) dataset IV. (a1)–(d1)–(a3)–(d3) Detection maps with low-rank loss, the detection maps without low-rank loss, and the corresponding ground-reference maps of anomalies, respectively.

on them in the training process to suppress the ability to reconstruct anomalies. However, in the experiments, we found that the generalization of this method is not sufficiently strong, as shown in Fig. 4(f1)–(f4). Therefore, the improvement of the weighted loss function for different pixels is an urgent problem.

Deep learning-based methods have a strong nonlinear fitting ability and show great potential in a variety of image-processing tasks. However, as shown in Table I, the calculation cost of deep learning-based methods should be considered, because it may be a problem in practical applications, especially when real-time detection is required. Additionally, as shown in Fig. 7(c), the stability of the network improves with an increase in the number of network layers. However, owing to the increase in the number of hyperparameters, hyperparameter adjustment is a problem that cannot be ignored. The lightweight network can not only reduce the operation time but also avoid tedious hyperparameter adjustment. Therefore, designing a lightweight network will be the focus of future work.

In this article, we propose a low-rank loss function for constraining the latent space. Low-rank loss considers the spatial attributes of the HSIs. Spatial attributes have been widely used in traditional algorithms (such as LRSMD-based methods), but they are rarely used in deep learning. In recent years, research works on the combination of traditional and deep learning-based algorithms have also received extensive attention. Traditional algorithms have greater interpretability, whereas deep learning-based algorithms can extract the deep features of images that have stronger representation ability. In future work, we will continue to seek ways to embed the spatial attributes of HSIs and traditional algorithms into deep learning-based models.

Finally, the scaled dot-product model is used as the attention-scoring function in the proposed RSA module.

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Several other common attention-scoring functions have been applied to calculate the weight matrix, such as the additive [46], concatenation [47], and bilinear models [48]. Compared with the above methods, the scaled dot-product model only needs to set the dimensions of **Q** and **K**, but the model is insensitive to them. An effective attention score function can provide more useful features for subsequent tasks. It is important to design an attention-scoring function suitable for HSI processing.

VI. Conclusion

In this article, we proposed an RSAAE for hyperspectral anomaly detection. RSAAE uses dense RSA modules to extract the main information of the HSIs to better reconstruct the background and suppress anomaly reconstruction. The proposed RSA module not only highlights the main information but also avoids the weakening of the original features. Additionally, we proposed a low-rank loss function for constraining the latent space, which improved the detection accuracy noticeably. Through experimental validation on four datasets, the proposed RSAAE demonstrated better performance than eight benchmark methods and was robust to hyperparameters.

References

[1] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5522214, doi: 10.1109/TGRS.2022.3144158.

[2] Z. Gong, W. Hu, X. Du, P. Zhong, and P. Hu, "Deep manifold embedding for hyperspectral image classification," *IEEE Trans. Cybern.*, vol. 52, no. 10, pp. 10430–10443, Oct. 2022, doi: 10.1109/TCYB.2021.3069790.

[3] M. Hu, C. Wu, and L. Zhang, "HyperNet: Self-supervised hyperspectral spatial–spectral feature understanding network for hyperspectral change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5543017, doi: 10.1109/TGRS.2022.3218795.

[4] X. Ou, L. Liu, B. Tu, G. Zhang, and Z. Xu, "A CNN framework with slow-fast band selection and feature fusion grouping for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5524716, doi: 10.1109/TGRS.2022.3156041.

[5] A. Zare, C. Jiao, and T. Glenn, "Discriminative multiple instance hyperspectral target characterization," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 10, pp. 2342–2354, Oct. 2018.

[6] X. Shang et al., "Target-constrained interference-minimized band selection for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 6044–6064, Jul. 2021, doi: 10.1109/TGRS.2020.3010826.

[7] J. Zhong, W. Xie, Y. Li, J. Lei, and Q. Du, "Characterization of background-anomaly separability with generative adversarial network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 6017–6028, Jul. 2021.

[8] J. Liu, Z. Hou, W. Li, R. Tao, D. Orlando, and H. Li, "Multipixel anomaly detection with unknown patterns for hyperspectral imagery," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 10, pp. 5557–5567, Oct. 2022, doi: 10.1109/TNNLS.2021.3071026.

[9] I. S. Reed and X. Yu, "Adaptive multiple-band CFAR detection of an optical pattern with unknown spectral distribution," *IEEE Trans. Acoust., Speech Signal Process.*, vol. 38, no. 10, pp. 1760–1770, Oct. 1990.

[10] H. Kwon, S. Z. Der, and N. M. Nasrabadi, "Adaptive anomaly detection using subspace separation for hyperspectral imagery," *Opt. Eng.*, vol. 42, no. 11, pp. 3342–3351, 2003.

[11] Q. Guo, B. Zhang, Q. Ran, L. Gao, J. Li, and A. Plaza, "Weighted-RXD and linear filter-based RXD: Improving background statistics estimation for anomaly detection in hyperspectral imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2351–2366, Jun. 2014.

[12] W.-M. Liu and C.-I. Chang, "Multiple-window anomaly detection for hyperspectral imagery," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jan. 2008, pp. 41–44.

[13] A. Schaum, "Joint subspace detection of hyperspectral targets," in *Proc. IEEE Aerosp. Conf.*, Mar. 2004, pp. 1818–1824.

[14] H. Kwon and N. M. Nasrabadi, "Kernel RX-algorithm: A nonlinear anomaly detector for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 2, pp. 388–397, Feb. 2005, doi: 10.1109/TGRS.2004.841487.

[15] J. Zhou, C. Kwan, B. Ayhan, and M. T. Eismann, "A novel cluster kernel RX algorithm for anomaly and change detection using hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 11, pp. 6497–6504, Nov. 2016, doi: 10.1109/TGRS.2016.2585495.

[16] Z. Yuan, H. Sun, K. Ji, Z. Li, and H. Zou, "Local sparsity divergence for hyperspectral anomaly detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 10, pp. 1697–1701, Oct. 2014.

[17] L. Wei and D. Qian, "Collaborative representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 3, pp. 1463–1474, Mar. 2015.

[18] Y. Xu, Z. Wu, J. Li, A. Plaza, and Z. Wei, "Anomaly detection in hyperspectral images based on low-rank and sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 4, pp. 1990–2000, Mar. 2016.

[19] Y. Zhang et al., "A low-rank and sparse matrix decomposition-based Mahalanobis distance method for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 3, pp. 1376–1389, Oct. 2015.

[20] Y. Qu et al., "Anomaly detection in hyperspectral images through spectral unmixing and low rank decomposition," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Beijing, China, Jul. 2016, pp. 1855–1858, doi: 10.1109/IGARSS.2016.7729476.

[21] Y. Qu et al., "Hyperspectral anomaly detection through spectral unmixing and dictionary-based low-rank decomposition," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4391–4405, Aug. 2018, doi: 10.1109/TGRS.2018.2818159.

[22] T. Cheng and B. Wang, "Graph and total variation regularized low-rank representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 391–406, Jan. 2020.

[23] T. Cheng and B. Wang, "Total variation and sparsity regularized decomposition model with union dictionary for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 2, pp. 1472–1486, Feb. 2021.

[24] S. Li, W. Wang, H. Qi, B. Ayhan, C. Kwan, and S. Vance, "Low-rank tensor decomposition based anomaly detection for hyperspectral imagery," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Quebec City, QC, Canada, Sep. 2015, pp. 4525–4529, doi: 10.1109/ICIP.2015.7351663.

[25] T. Jiang, W. Xie, Y. Li, J. Lei, and Q. Du, "Weakly supervised discriminative learning with spectral constrained generative adversarial network for hyperspectral anomaly detection," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 11, pp. 6504–6517, Nov. 2022, doi: 10.1109/TNNLS.2021.3082158.

[26] W. Xie, B. Liu, Y. Li, J. Lei, and Q. Du, "Autoencoder and adversarial-learning-based semisupervised background estimation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 8, pp. 5416–5427, Aug. 2020.

[27] Y. Li, T. Jiang, W. Xie, J. Lei, and Q. Du, "Sparse coding-inspired GAN for hyperspectral anomaly detection in weakly supervised learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5512811, doi: 10.1109/TGRS.2021.3102048.

[28] K. Jiang et al., "E2E-LIADE: End-to-end local invariant autoencoding density estimation model for anomaly target detection in hyperspectral image," *IEEE Trans. Cybern.*, vol. 52, no. 11, pp. 11385–11396, Nov. 2022, doi: 10.1109/TCYB.2021.3079247.

[29] S. Wang, X. Wang, L. Zhang, and Y. Zhong, "Auto-AD: Autonomous hyperspectral anomaly detection network based on fully convolutional autoencoder," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5503314, doi: 10.1109/TGRS.2021.3057721.

[30] G. Fan, Y. Ma, X. Mei, F. Fan, J. Huang, and J. Ma, "Hyperspectral anomaly detection with robust graph autoencoders," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5511314, doi: 10.1109/TGRS.2021.3097097.

[31] Y. Liu, W. Xie, Y. Li, Z. Li, and Q. Du, "Dual-frequency autoencoder for anomaly detection in transformed hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5523613, doi: 10.1109/TGRS.2022.3152263.

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[32] C. K. I. Williams and D. Barber, "Bayesian classification with Gaussian processes," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 20, no. 12, pp. 1342–1351, Dec. 1998.
[33] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[34] S. Boyd, N. Parikh, E. Chu, B. Peleato, and J. Eckstein, "Distributed optimization and statistical learning via the alternating direction method of multipliers," *Found. Trends Mach. Learn.*, vol. 3, no. 1, pp. 1–122, Jan. 2011.
[35] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, and A. N. Gomez, "Attention is all you need," 2017, *arXiv: 1706.03762*.
[36] X. Zhang et al., "Spectral–spatial self-attention networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5512115.
[37] H. Dong, W. Ma, L. Jiao, F. Liu, and L. Li, "A multiscale self-attention deep clustering for change detection in SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5207016.
[38] M. Mao, H. Wang, P. Nie, S. Xiao, and R. Wu, "DenoiseNet: Deep generator and discriminator learning network with self-attention applied to ocean data," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5922212, doi: 10.1109/TGRS.2022.3217402.
[39] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," in *Proc. Int. Conf. Mach. Learn.*, vol. 37, Jul. 2015, pp. 448–456.
[40] S. M. Siniscalchi and V. M. Salerno, "Adaptation to new microphones using artificial neural networks with trainable activation functions," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 28, no. 8, pp. 1959–1965, Aug. 2017.
[41] R. Girshick, "Fast R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1440–1448.
[42] J. Kerekes, "Receiver operating characteristic curve confidence intervals and regions," *IEEE Geosci. Remote Sens. Lett.*, vol. 5, no. 2, pp. 251–255, Apr. 2008.
[43] M. Song, X. Shang, and C.-I. Chang, "3-D receiver operating characteristic analysis for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 11, pp. 8093–8115, Nov. 2020.
[44] C. Ferri, J. Hernandez-Orallo, and P. Flach, "A coherent interpretation of AUC as a measure of aggregated classification performance," in *Proc. Int. Conf. Mach. Learn.*, 2011, pp. 657–664.
[45] Z. Wu et al., "Hyperspectral anomaly detection with relaxed collaborative representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5533417, doi: 10.1109/TGRS.2022.3190327.
[46] C. Wu, F. Wu, T. Qi, Y. Huang, and X. Xie, "FastFormer: Additive attention can be all you need," 2021, *arXiv:2108.09084*.
[47] C. Tan, F. Wei, W. Wang, W. Lv, and M. Zhou, "Multiway attention networks for modeling sentence pairs," in *Proc. 27th Int. Joint Conf. Artif. Intell.*, Jul. 2018, pp. 4411–4417.
[48] J. H. Kim, J. Jun, and B. T. Zhang, "Bilinear attention networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2018, pp. 1–11.



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